Niloofar Asefi

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Ph.D. candidate in Electrical and Computer Engineering at UC Santa Cruz specializing in fundamental generative AI for scientific data (climate, weather, and ocean modeling). Experienced in designing, training, and evaluating large-scale generative models and RL-based control — including **physics-informed ML**, **diffusion models**, **transformers, reinforcement learning**, and **neural operators**— applied to conditional 3D generation, multimodal conditioning, high-resolution reconstruction from sparse, noisy observations, and climate-intervention control. Interdisciplinary background spanning dynamical systems and chaos, turbulence physics, materials science, and deep learning. Fluent in PyTorch/JAX and multi-GPU distributed training on national HPC systems. 2 years mentoring student researchers.

EXPERIENCE

Fujitsu Research of America, Inc. June 2026 – present

ML Research Intern — research on generative AI, physics-informed ML models, diffusion models

- Project complete; **first-author** abstract submitted to **AGU 2026** and a **first-author journal manuscript in preparation**.
- Studied **out-of-distribution generalization** of a conditional diffusion model across physical regimes: transferred a depth-aware DDPM trained on the Gulf of Mexico to the Kuroshio western boundary current—a substantially harder target distribution with sharp thermal/velocity fronts, energetic mesoscale eddies, and strong bathymetric control.
- Designed **physics-guided loss terms** (spatial-gradient, kinetic-energy, and strain-weighted MSE) for the DDPM training objective, improving physical fidelity of reconstructed ocean states.
- Built a multi-axis evaluation suite that diagnoses model mechanism rather than only scoring accuracy: spectral analysis, gradient diagnostics, velocity-derived strain rate and relative vorticity, and ensemble-based uncertainty estimates.

SELECTED PUBLICATIONS

1. Asefi, N., et al. *"High-Resolution Probabilistic Estimation of Three-Dimensional Regional Ocean Dynamics from Sparse Surface Observations."* Under revision **(2026).** arXiv:2604.02850.

2. Asefi, N., et al. *"Generative Lagrangian Data Assimilation for Ocean Dynamics under Extreme Sparsity."* **Machine Learning: Earth (2025)**.

3. Asefi, N., et al., *"Photocatalytic Properties of $BiFeO_3$ Powders Synthesized by the Mixture of CTAB and Glycine," Scientific Reports, vol. 13, no. 1, 2023.*

Plus **3 additional first-author publications** in solution-combustion synthesis of $BiFeO_3$ (*Materials Chemistry and Physics* 2019; *J. Electronic Materials* 2019; *J. Sol-Gel Science and Technology* 2018).

PROJECT EXPERIENCE

Conditional Diffusion for 3D Generation from Sparse Observations 2025 – 2026

- Developed a novel **depth-aware diffusion architecture (DDPM)** trained on multi-dimensional PDE-based simulation outputs to reconstruct **high-resolution 3D fields** conditioned on sparse 2D surface observations — a conditional generation task that recovers full 3D ocean states from extremely limited surface data.
- Built and processed large-scale satellite and reanalysis datasets into scalable training pipelines; **achieved O(1) memory complexity in vertical resolution**, enabling high-resolution 3D inference at scale.

Reinforcement Learning for Climate Intervention Strategies 2026 – Present

- Built REINFORCE and actor-critic controllers over a continuous 5-zonal-band action space, coupled to two environments: the LUCIE climate emulator (trained on ERA5 reanalysis) and a JAX-GCM differentiable Earth-system model.
- Designed and compared multiple reward formulations to study how reward specification shapes the learned control policy.

Generative Sparse-to-Dense Reconstruction with Diffusion Models & Neural Operators 2025

- Developed generative ML approaches, including diffusion models and **Fourier Neural Operators (FNO),** as alternatives to traditional data assimilation methods such as ensemble Kalman filtering and variational techniques.

- Designed and evaluated multi-source conditional generation frameworks fusing heterogeneous inputs, demonstrating multimodal conditioning strategies for controllable generative models.
- Built fully reproducible, large-scale training and evaluation pipelines, documented experiments for team reproducibility.

Explainable AI: Clustered Compositional Explanations (CCE) for NLP 2024

- Studied and analyzed neuron-level compositional representations in LSTM and Transformer-based NLP models.
- Contributed to model interpretability research, gaining experience with NLP architectures and evaluation methodologies.

EDUCATION

Ph.D. Candidate, Electrical and Computer Engineering 2022 – Present

University of California, Santa Cruz (UCSC), **Advisor: Prof. Ashesh Chattopadhyay**

M.S., Materials Engineering 2016 – 2019

Iran University of Science and Technology (IUST)

MENTORSHIP & TEACHING

Mentor, Science Internship Program (SIP), UC Santa Cruz Summer 2024, Summer 2025

- **2025:** Supervised high school researchers on a project reconstructing sea surface height field over the Gulf of Mexico with U-Net and CNN models under systematically varied observational sparsity—a hands-on generalization study. Guided the full arc: data preprocessing, training, and evaluation.
- **2024:** Mentored students writing survey papers and doing coding experiments on CNN and ResNet architectures for cancer detection.

Teaching Assistant, UC Santa Cruz: Calculus for Engineering & Mathematics; Python; Signals & Systems.

SKILLS

- **Languages & Tools:** Python (primary), SQL, NumPy, MATLAB, Scikit-learn, Git
- **ML Frameworks:** PyTorch (DDP/FSDP/ZeRO), JAX, multi-GPU training, scalable ML pipelines
- **AI/ML: Diffusion models (DDPM, DDIM)**, conditional generative modeling, multimodal conditioning**, physics-informed ML**, **surrogate modeling**, **reinforcement learning**, actor-critic methods, uncertainty quantification
- **Architectures:** Attention-UNet, **Fourier Neural Operators (FNO)**, Transformers, multimodal conditioning
- **MLOps & Systems:** Reproducible training pipelines, model evaluation, HPC workloads (Derecho, Bridges2 — PBS/Slurm), distributed training, geospatial raster data processing

AWARDS & FELLOWSHIPS

NCC26 Travel Award, SIAM Conference (2026), **Dissertation-Year Fellowship**, UCSC (2026–27) · **Best Presentation Award**, 22nd Annual UCSC Graduate Research Symposium (2026) · **Mayfield Scholar**, UCSC (2026) · **Graduate Dean's Travel Award**, UCSC (2025–26) · **Graduate Regents Fellowship**, UCSC (2022–23) · **Graduate International Department Fellowship**, UCSC (2022–23) · **Research Publication Award**, IUST (2019)

CONFERENCES & PRESENTATIONS

- **American Geophysical Union (AGU25**), Advancing Climate Science with Deep Learning II, Dec. 2025
- **SIAM Northern California Conference (NCC25)**, Oct. 2025
- **American Geophysical Union (AGU26)**, accepted for oral and poster presentation, Dec. 2026 (upcoming)
- **SIAM Northern California Conference (NCC26)**, accepted, Oct. 2026 (upcoming)